Title

Emotion Recognition from Social Media Text Using Large Language Models

Abstract

Social media platforms generate a large amount of text every day, where people express their thoughts, opinions, and emotions. Understanding these emotions can help organizations, businesses, and researchers analyze public opinion, improve customer satisfaction, and monitor mental well-being. Traditional sentiment analysis usually identifies text as positive, negative, or neutral, but it cannot accurately recognize specific emotions such as joy, sadness, anger, fear, surprise, or love.

This project proposes an **Emotion Recognition from Social Media Text Using Large Language Models (LLMs)** framework to automatically detect emotions from social media posts. The system uses advanced Transformer-based language models such as **BERT, DistilBERT, or RoBERTa**, which are pre-trained on large text datasets and fine-tuned for emotion classification. The project includes text preprocessing, tokenization, text normalization, feature extraction, and emotion prediction using benchmark datasets such as **GoEmotions** or other publicly available emotion datasets.

The proposed system classifies each social media post into different emotion categories, including **joy, sadness, anger, fear, surprise, and love**. The performance of the model is evaluated using metrics such as **accuracy, precision, recall, F1-score, and confusion matrix** to measure the effectiveness of the emotion classification model.

The developed system aims to provide an accurate and efficient solution for emotion recognition from social media text. It can be applied in areas such as **social media monitoring, customer feedback analysis, mental health awareness, public opinion analysis, and business decision-making**.

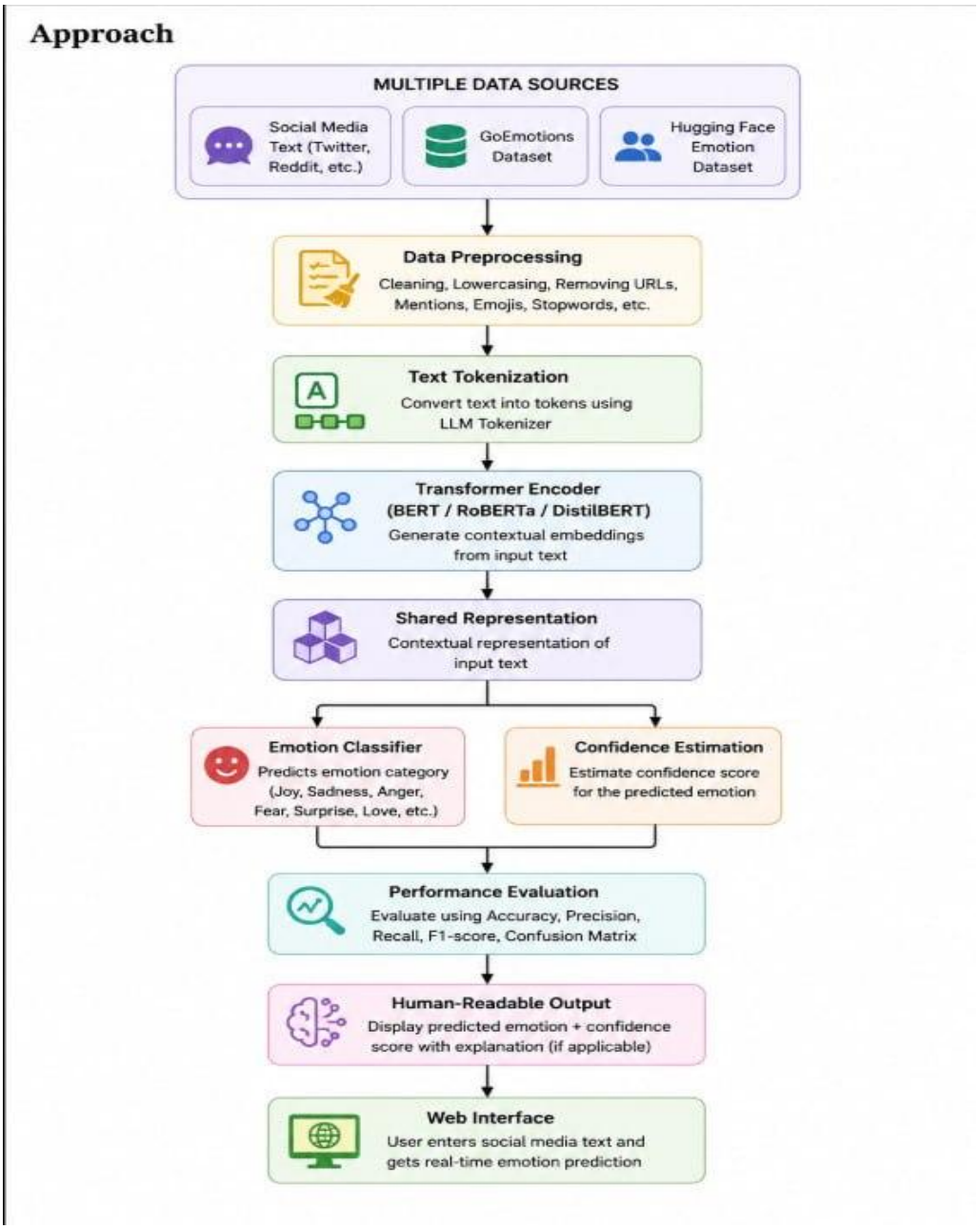
Datasets

Dataset	Dataset URL Link
GoEmotions Dataset	https://huggingface.co/datasets/google-research-datasets/go_emotions
Emotion Dataset (Kaggle)	https://www.kaggle.com/datasets/praveengovi/emotions-dataset-for-nlp

Approach

1. **Data Collection**
    - Collect social media text data from the GoEmotions dataset or Emotion Dataset (Kaggle).
2. **Data Preprocessing**
    - Remove unwanted symbols, URLs, and special characters.
    - Convert text to lowercase.
    - Perform tokenization, stop-word removal, stemming, and lemmatization.
3. **Text Normalization**
    - Expand contractions and normalize the text for better model understanding.
4. **Feature Extraction**
    - Convert the processed text into numerical representations using Transformer tokenizers.
5. **Model Training**
    - Fine-tune a pre-trained Large Language Model such as **BERT**, **DistilBERT**, or **RoBERTa** for emotion classification.

6. **Emotion Classification**
    - Predict emotions such as **Joy, Sadness, Anger, Fear, Surprise, Love, and Neutral** from the input text.
7. **Model Evaluation**
    - Evaluate the model using:
        - Accuracy
        - Precision
        - Recall
        - F1-Score
        - Confusion Matrix
8. **Result Visualization**
    - Display the detected emotion along with confidence scores.
    - Visualize the distribution of emotions using graphs or charts.



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